# SG No. 35 $C_{2v}^{11}$ $Cmm2$ [ orthorhombic ]

* plus set: $+[0,\,0,\,0]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$
* Wyckoff site: `2a`, site symmetry: `mm2`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `mm2`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0,\,0,\,Z]$ | $[0,\,0,\,z]$ | `[1,2,3,4]` |

Table 2: Wyckoff bond: `2b@2a`, bond symmetry: `mm2`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0,\,Y,\,0]$ | $[0,\,0,\,z]$ | `[1,-2,-3,4]` |

Table 3: Wyckoff bond: `2c@2a`, bond symmetry: `mm2`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X,\,0,\,0]$ | $[0,\,0,\,z]$ | `[1,-2,3,-4]` |

Table 4: Wyckoff bond: `4d@2a`, bond symmetry: `..2`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X,\,Y,\,0]$ | $[0,\,0,\,z]$ | `[1,-2]` |
| 2 | $[X,\,-Y,\,0]$ | $[0,\,0,\,z]$ | `[3,-4]` |

Table 5: Wyckoff bond: `4e@2a`, bond symmetry: `.m.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X,\,0,\,Z]$ | $[0,\,0,\,z]$ | `[1,3]` |
| 2 | $[-X,\,0,\,Z]$ | $[0,\,0,\,z]$ | `[2,4]` |

Table 6: Wyckoff bond: `4f@2a`, bond symmetry: `m..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0,\,Y,\,Z]$ | $[0,\,0,\,z]$ | `[1,4]` |
| 2 | $[0,\,-Y,\,Z]$ | $[0,\,0,\,z]$ | `[2,3]` |

Table 7: Wyckoff bond: `8g@2a`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X,\,Y,\,Z]$ | $[0,\,0,\,z]$ | `[1]` |
| 2 | $[-X,\,-Y,\,Z]$ | $[0,\,0,\,z]$ | `[2]` |
| 3 | $[X,\,-Y,\,Z]$ | $[0,\,0,\,z]$ | `[3]` |

*continued ...*

Table 7

| No. | vector | center | mapping |
|---|---|---|---|
| 4 | $[-X, Y, Z]$ | $[0, 0, z]$ | `[4]` |

* Wyckoff site: `2b`, site symmetry: `mm2`

Table 8: Wyckoff bond: `2a@2b`, bond symmetry: `mm2`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, 0, Z]$ | $\left[0, \frac{1}{2}, z\right]$ | `[1,2,3,4]` |

Table 9: Wyckoff bond: `2b@2b`, bond symmetry: `mm2`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, Y, 0]$ | $\left[0, \frac{1}{2}, z\right]$ | `[1,-2,-3,4]` |

Table 10: Wyckoff bond: `2c@2b`, bond symmetry: `mm2`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, 0, 0]$ | $\left[0, \frac{1}{2}, z\right]$ | `[1,-2,3,-4]` |

Table 11: Wyckoff bond: `4d@2b`, bond symmetry: `..2`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, 0]$ | $\left[0, \frac{1}{2}, z\right]$ | `[1,-2]` |
| 2 | $[X, -Y, 0]$ | $\left[0, \frac{1}{2}, z\right]$ | `[3,-4]` |

Table 12: Wyckoff bond: `4e@2b`, bond symmetry: `.m.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, 0, Z]$ | $\left[0, \frac{1}{2}, z\right]$ | `[1,3]` |
| 2 | $[-X, 0, Z]$ | $\left[0, \frac{1}{2}, z\right]$ | `[2,4]` |

Table 13: Wyckoff bond: `4f@2b`, bond symmetry: `m..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, Y, Z]$ | $\left[0, \frac{1}{2}, z\right]$ | `[1,4]` |
| 2 | $[0, -Y, Z]$ | $\left[0, \frac{1}{2}, z\right]$ | `[2,3]` |

Table 14: Wyckoff bond: `8g@2b`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $\left[0, \frac{1}{2}, z\right]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $\left[0, \frac{1}{2}, z\right]$ | `[2]` |
| 3 | $[X, -Y, Z]$ | $\left[0, \frac{1}{2}, z\right]$ | `[3]` |
| 4 | $[-X, Y, Z]$ | $\left[0, \frac{1}{2}, z\right]$ | `[4]` |

* Wyckoff site: `4c`, site symmetry: `..2`

Table 15: Wyckoff bond: `4a@4c`, bond symmetry: `..2`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, z\right]$ | `[1,-2]` |
| 2 | $[X, -Y, 0]$ | $\left[\frac{1}{4}, \frac{3}{4}, z\right]$ | `[3,-4]` |

Table 16: Wyckoff bond: `4b@4c`, bond symmetry: `..2`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, 0, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, z\right]$ | `[1,2]` |
| 2 | $[0, 0, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, z\right]$ | `[3,4]` |

Table 17: Wyckoff bond: `8c@4c`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, z\right]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $\left[\frac{1}{4}, \frac{1}{4}, z\right]$ | `[2]` |
| 3 | $[X, -Y, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, z\right]$ | `[3]` |
| 4 | $[-X, Y, Z]$ | $\left[\frac{1}{4}, \frac{3}{4}, z\right]$ | `[4]` |

* Wyckoff site: `4d`, site symmetry: `.m.`

Table 18: Wyckoff bond: `4a@4d`, bond symmetry: `.m.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, 0, Z]$ | $[x, 0, z]$ | `[1,3]` |
| 2 | $[-X, 0, Z]$ | $[-x, 0, z]$ | `[2,4]` |

Table 19: Wyckoff bond: `4b@4d`, bond symmetry: `.m.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, Y, 0]$ | $[x, 0, z]$ | `[1,-3]` |
| 2 | $[0, -Y, 0]$ | $[-x, 0, z]$ | `[2,-4]` |

Table 20: Wyckoff bond: `8c@4d`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[x, 0, z]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $[-x, 0, z]$ | `[2]` |
| 3 | $[X, -Y, Z]$ | $[x, 0, z]$ | `[3]` |
| 4 | $[-X, Y, Z]$ | $[-x, 0, z]$ | `[4]` |

* Wyckoff site: `4e`, site symmetry: `m..`

Table 21: Wyckoff bond: `4a@4e`, bond symmetry: `m..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, Y, Z]$ | $[0, y, z]$ | `[1,4]` |
| 2 | $[0, -Y, Z]$ | $[0, -y, z]$ | `[2,3]` |

Table 22: Wyckoff bond: `4b@4e`, bond symmetry: `m..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, 0, 0]$ | $[0, y, z]$ | `[1,-4]` |
| 2 | $[-X, 0, 0]$ | $[0, -y, z]$ | `[2,-3]` |

Table 23: Wyckoff bond: `8c@4e`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[0, y, z]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $[0, -y, z]$ | `[2]` |
| 3 | $[X, -Y, Z]$ | $[0, -y, z]$ | `[3]` |
| 4 | $[-X, Y, Z]$ | $[0, y, z]$ | `[4]` |

* Wyckoff site: `8f`, site symmetry: `1`

Table 24: Wyckoff bond: `8a@8f`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[x, y, z]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $[-x, -y, z]$ | `[2]` |
| 3 | $[X, -Y, Z]$ | $[x, -y, z]$ | `[3]` |
| 4 | $[-X, Y, Z]$ | $[-x, y, z]$ | `[4]` |